AI-Lingo

Contextual Language Immersion App

Project Planning & Management
Project Proposal and Project Plan

1. Project Proposal

1.1 Project Overview

AI-Lingo is an AI-powered English learning mobile application. Users practice realistic conversations with Gemini based on scenarios such as job interviews, restaurants, travel, and business meetings. The app adapts practice to the learner's English level and provides corrections, vocabulary suggestions, evaluation, and progress tracking.

1.2 Problem Statement

Learners may struggle to find realistic, personalized English practice and timely feedback. AI-Lingo addresses this by combining scenario-based AI conversations, feedback, vocabulary review, and learning progress in one mobile application.

1.3 Project Objectives

- Provide personalized English practice through AI conversations.
- Adapt dialogue difficulty to the user's selected or assessed level.
- Apply prompt engineering, prompt chaining, and constrained prompts.
- Evaluate grammar, vocabulary, and conversation performance.
- Save and review vocabulary and track progress.
- Use Hive for local vocabulary and selected cached learning data.

1.4 Project Scope and Technology Stack

The MVP is one Flutter application for end users, with no separate admin app. It includes authentication, level selection/assessment, scenario selection, text-based AI conversation, evaluation, vocabulary, and progress tracking. Flutter/Dart and Bloc/Cubit handle the client and state; Firebase Authentication and Cloud Firestore handle accounts and cloud data; Firebase AI Logic connects to Gemini; Hive provides local NoSQL storage. Voice conversations and advanced gamification are future enhancements.

1.5 Expected Outcome

A working mobile app that helps learners practice English, understand mistakes, save useful vocabulary, and monitor progress. It demonstrates mobile development, Firebase integration, AI integration, prompt engineering, local storage, security, and testing.

2. Project Plan

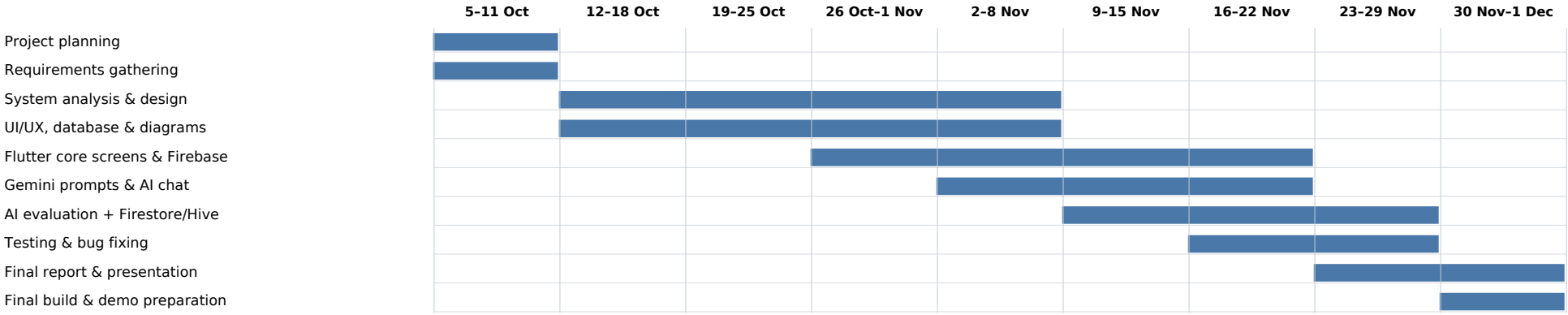
2.1 Development Phases, Milestones, and Deliverables

Work package	Planned dates	Main activities	Deliverable / milestone
Project proposal and planning	5–12 Oct	Finalize overview, objectives, scope, project plan, and Gantt chart.	Planning document completed
Requirements gathering	5–12 Oct	Define target users, user stories/use cases, functional and non-functional requirements.	Requirements document completed
System analysis & design	13 Oct–6 Nov	Architecture, use case diagram, data model, data flow/sequence diagrams, UI/UX wireframes.	System analysis and design package
Implementation	7–24 Nov	Build Flutter screens, Firebase authentication/Firestore, Gemini chat/evaluation, Hive storage, and progress.	Working application and source code
Testing & bug fixing	20–27 Nov	Run unit, widget, and integration tests; verify security, error handling, and offline behavior.	Test results and critical fixes
Final report & presentation	25–29 Nov	Complete technical documentation, user guide, presentation, and demo script.	Final report and presentation ready
Final build & buffer	30 Nov–1 Dec	Prepare final Android build, verify demo, check submission files, and resolve last critical issues.	Final project package ready by 1 Dec

2.2 Gantt Chart — Revised Schedule (5 October–1 December 2026)

Project planning and requirements gathering run from 5 October through 12 October 2026. The Literature Review is excluded because it will be handled by the reviewers. All planned project work finishes by 1 December 2026, leaving 2–4 December as a contingency buffer before the final deadline.

AI-Lingo — Gantt Chart (5 Oct–1 Dec 2026)



2.3 Resource Allocation

Resources include the student team, laptops, internet access, Flutter/Dart tools, Firebase services, Gemini through Firebase AI Logic, Hive, Git/GitHub, and a UI/UX design tool. A separate Node.js backend is not included in the planned architecture.

2.4 Progress Monitoring and Risk Controls

Track tasks using GitHub Issues or a task board and review progress at each milestone. Main risks include late implementation, Firebase/Gemini quota limits, AI output quality, and integration bugs. Mitigate them by validating Gemini/Firebase integration early, keeping the MVP focused, testing continuously, and reserving 30 November–1 December for final build checks.

Planning note: Planning and requirements are scheduled from 5–12 October. The overall project targets completion on 1 December 2026.